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Class : 10

Register Number 1 0 2 1 1

COMMON HALF YEARLY EXAMINATION, 2025 - 26

SCIENCE

Time Allowed : 3.00 Hours] [Max. Marks : 75

PART – I

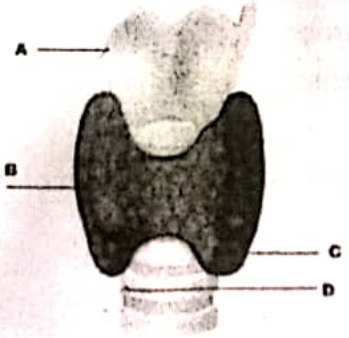
I. **Answer to all the questions** 12X1=12

1. Newton's III law is applicable
   a) for a body is at rest b) for a body in motion
   c) both a & b d) only for bodies with equal masses
2. In a simple circuit, why does the bulb glow when you close the switch?
   a) The switch produces electricity b) Closing the switch completes the circuit.
   c) Closing the switch breaks the circuit d) The bulb is getting charged.
3. which of the following isotope can be cure Anemia
   a) Iron - 59 b) Sodium - 24 c) Iodine - 131 d) Cobalt - 60
4. which of the following group contains the member of halogen family.
   a) 17th b) 15th c) 18th d) 16th
5. Solubility of NaCl in 100 ml water is 36 g. If 25 g of salt is dissolved in 100 ml of water how much more salt is required for saturation ----------
   a) 12 g b) 11g c) 16g d) 20g
6. which of the following factor influencing the rate of a reaction
   a) Temperature b) Catalyst c) Pressure d) all the above
7. Which is formed during anaerobic respiration
   a) Carbohydrate b) Ethyl alcohol b) Acetyl CoA d) Pyruvate
8. which one of the following blood cells number increases during conditions of **allergy**
   a) Neutrophils b) Eosinophils c) Basophils d) Lymphocytes
9. Node of Ranvier is found in
   a) muscles b) axons c) dendrites d) cyton
10. Which one of the following hormones is naturally not found in plants:
    a) 2, 4-D b) GA3 c) Gibberellin d) IAA
11. Excessive consumption of alcohol leads to
    a) Loss of memory b) Cirrhosis of liver
    c) State of hallucination d) Supression of brain function
12. A renewable source of energy is
    a) petroleum b) coal c) nuclear fuel d) solar energy

PART – II

II. **Answer to any seven questions.** 7X2=14
(question no. 22 is compulsory)

13. State Ohm's law.
14. Name three animals, which can hear ultrasonic vibrations
15. What is Molar volume of a gas?
16. Differentiate reversible and irreversible reactions
17. Name the parts of the hind brain.
18. **Identify the parts A, B, C and D**



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19. How can you determine the age of the fossils?
20. State the applications of DNA fingerprinting technique.
21. What is Sprite?
22. A charge of 12 coulomb flows through a bulb in 5 second. What is the current through the bulb?

PART – III

III. Answer to any seven questions. (question no. 32 is compulsory) **7X4=28**

23. Give any four of the applications of universal law gravitation.
24. i) Define one calorie.
    ii) What is co-efficient of apparant expansion?
25. i) Give any two uses of radio isotopes in the field of agriculture?
    ii) a) The energy released in a nuclear fission process is about ---
    b) Name of The nuclear bomb that was dropped in Hiroshima during World War II ---
26. i) **Match the following**

a) Galvanisation	-	Noble gas elements
b) Calcination	-	Coating with Zn
c) Dental filling	-	Heating in the absence of air
d) Group 18 elements	-	Silver-tin amalgam

    ii) Give the equation for formation of rust.
27. i) What happens when $MgSO_4.7H_2O$ is heated? Write the appropriate equation
    ii) Define solubility
28. Describe the structure of spinal cord
29. i) Write the characteristics of insect pollinated flowers.
    ii) What is colostrum?
30. Explain the structure of a chromosome.
31. Discuss the importance of biotechnology in the field of medicine.
32. i) calculate gram mass of 2 moles of hydrogen molecule
    ii) Calculate the % of oxygen in $Al_2(SO_4)_3$.
    (Atomic mass: (Al - 27, O -16, S - 32)

PART – IV

IV. Answer to all the questions. **3X7= 21**

33. (a) i) List any five properties of light
    ii) Why does the sky appear in blue colour?

(OR)

(b) i) What is a nuclear reactor? Explain its essential parts with their functions.
    ii) Write any two elements which are used for inducing radioactivity?

34. (a) i) Write notes on various factors affecting solubility.
    ii) The aquatic animals live more in cold region Why?

(OR)

(b) i) Arrive at, systematically, the IUPAC name of the compound: $CH_3–CH_2–CH_2–OH$.
    ii) Differentiate soaps and detergents

35. (a) i) Differentiate Aerobic and Anaerobic respiration
    ii) List out any three of the parasitic adaptations in leech

(OR)

(b) i) What are the various routes by which transmission of HIV takes place ?
    ii) How are e-wastes generated?
    iii) list out 3R Approach.

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